

# Gtm T2

Verified delivered configuration · interactive mesh available in [report.html](#)

**356 km/h**

DASH SPEED

**0.58 h**

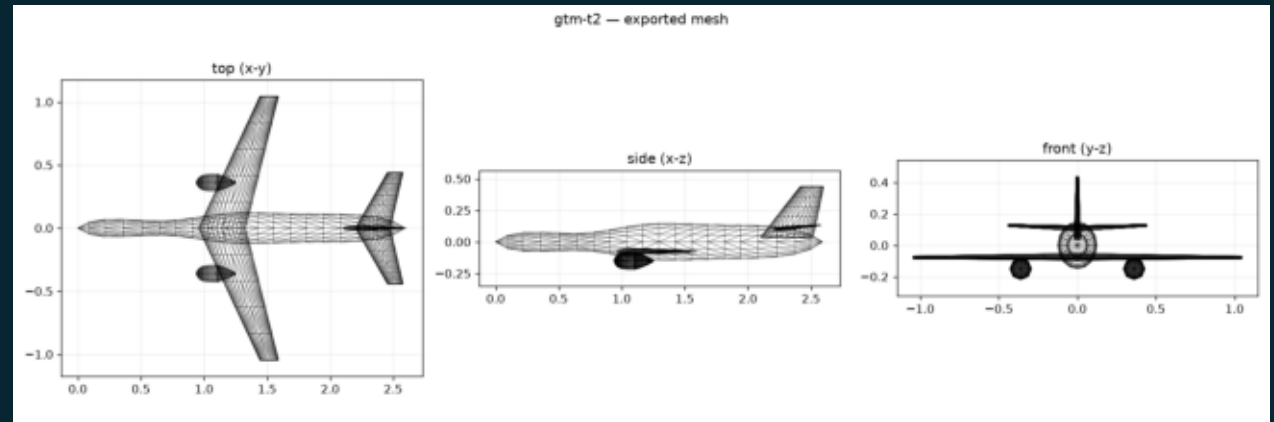
ENDURANCE

**0.0 kg**

PAYLOAD

**26.2 kg**

MTOW



# Flight-worthiness gate review

## ● Schema

source=yes; optimized=yes; engine requirement=yes; payload requirement=yes; re

## ● Geometry truth

read-back=yes; bbox=yes; mesh checks=28

## ● Packing

fuel-volume margin 0.058 m<sup>3</sup>; engine/payload clear

## ● Balance

component-model SM full 0.336; reserve 0.351 MAC

## ● Pitch trim

tail incidence -5.055° vs spec -5.057°; CM residual 0.0000

## ● Stall

Vstall 27.677 m/s ≤ 31.0 m/s at CLmax 1.20

## ● Directional authority

cant-corrected fin volume Vv 0.1163 ≥ 0.02 (source-locked reproduction)

## ● Structures

+4g failure index -0.882; CL 0.791/0.791; lift closed=yes; mass closed=yes; ti

## ● Endurance & thrust

0.58 h; cruise/dash thrust ≥ buildup drag

## ● Reproduction closure honesty

feasible=yes; driver=yes; post-delivery verify=yes; calibration=yes; frozen=ye

## ● Shape fidelity

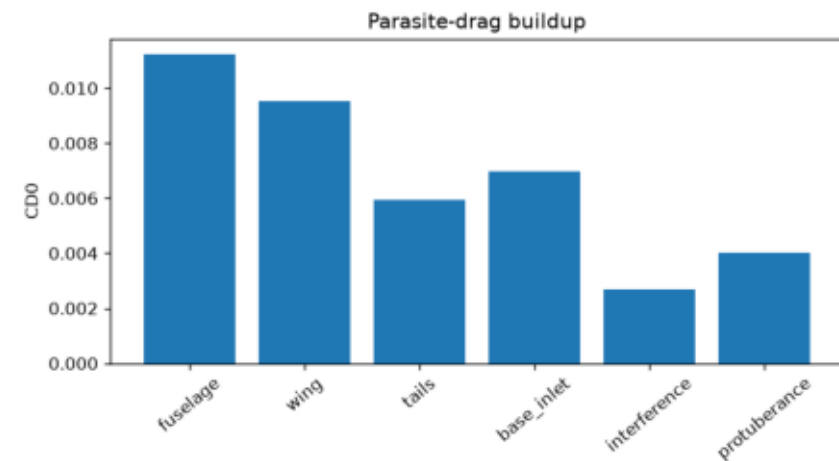
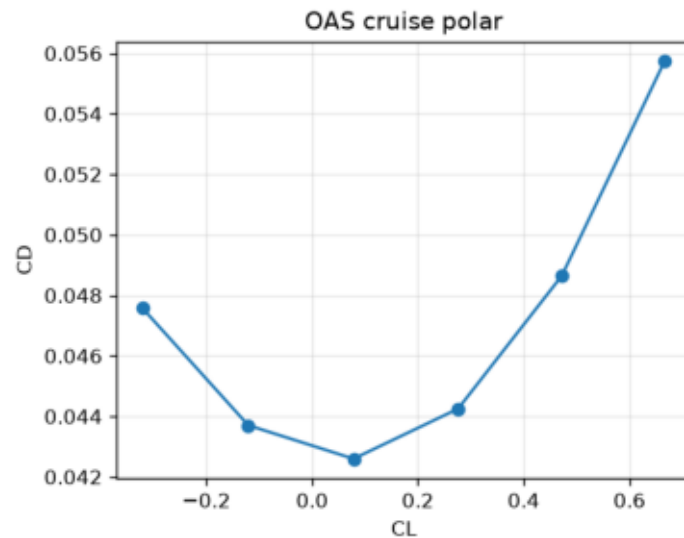
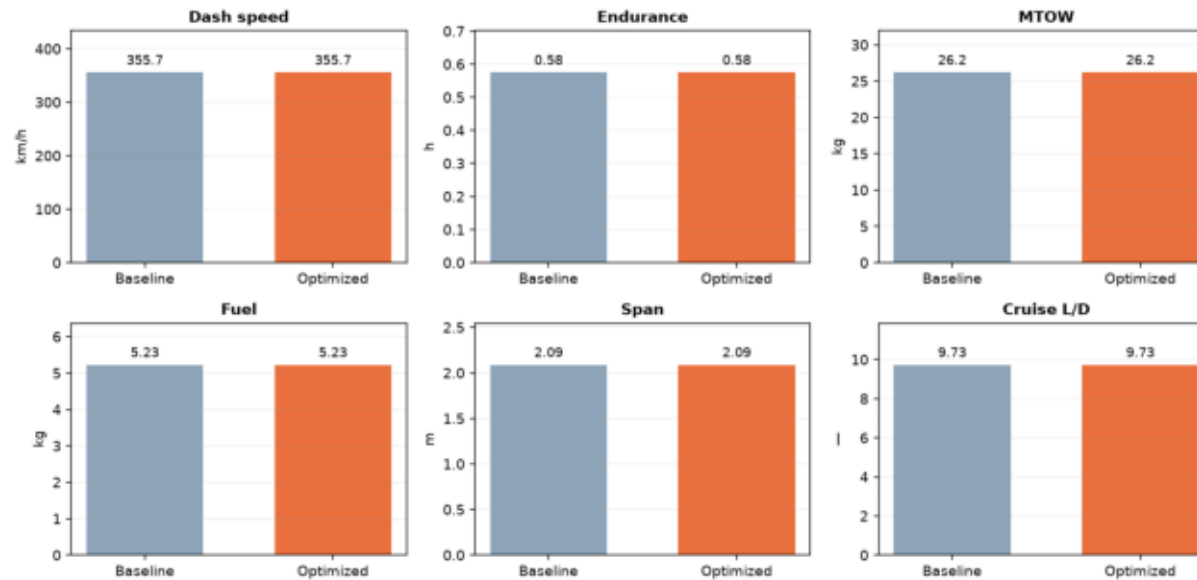
span/L 0.81; root/L 0.15; sweep 25.0°; 0 scored shape departures; 0 total audi

## ● Cross-checks & traceability

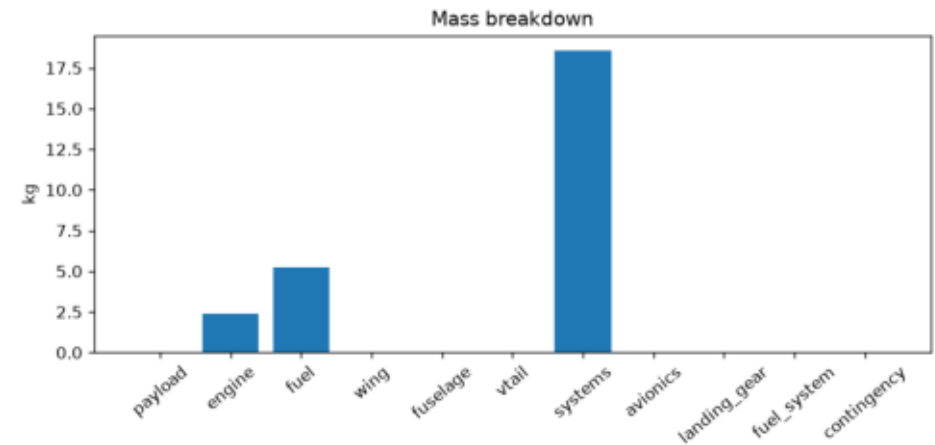
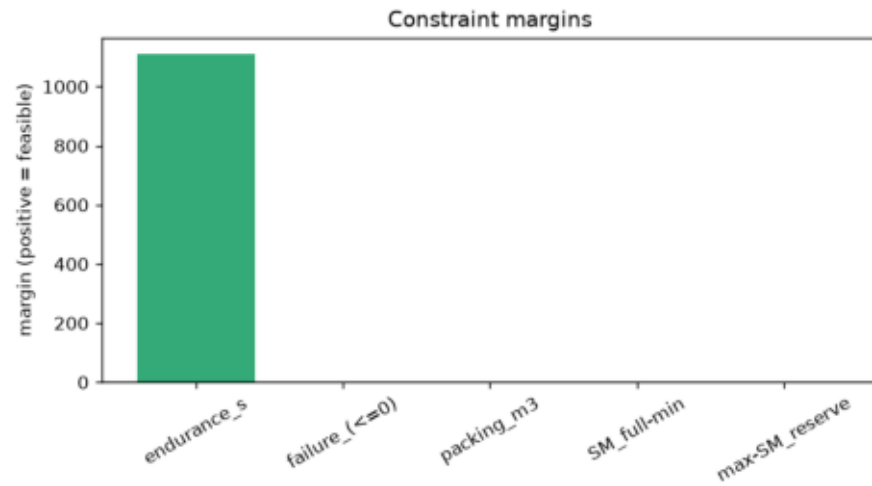
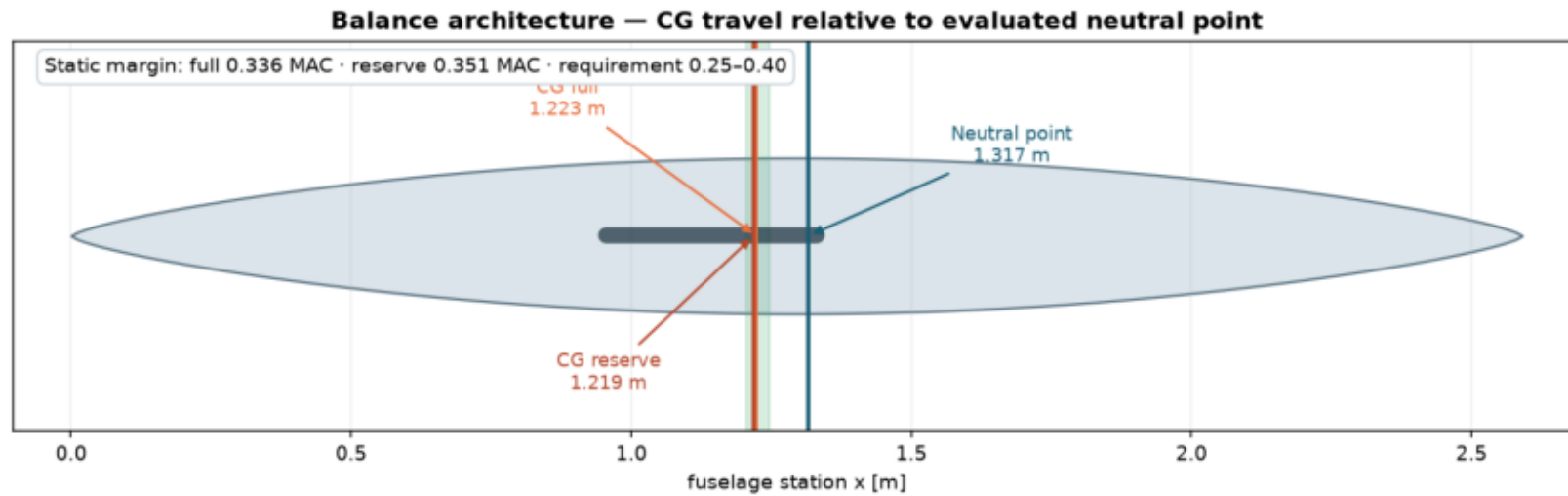
core 12/12; VSPAERO/OAS CL<sub>α</sub> ratio 0.950; NP method spread 0.198 MAC (diagnosti

# Performance and aerodynamic evidence

Design evolution: source baseline → verified optimum



# Balance and structural margins



# Verification summary and engineering limits

**Validation core: 12/12 · all recorded checks: 14/14**

- ✓ k450 tsfc kg per kgf hr
- ✓ breguet round trip
- ✓ isa T sl
- ✓ isa rho sl
- ✓ oas induced drag vs elliptic
- ✓ cantilever deflection identity
- ✓ thin airfoil cm ac
- ✓ fin volume coefficient
- ✓ fin te within body
- ✓ hybrid component provenance
- ✓ wing mass buildup vs oas
- ✓ vspaero vs oas CL
- ✓ tacs artifact schema

## Documented limitations

- Engine thrust-lapse and part-throttle TSFC are assumptions anchored by one static vendor point.
- Stall uses an assumed  $CL_{max}=1.20$ ; no high-lift or nonlinear separation model is present.
- OAS/VSPAERO are lifting-surface models; fins are gated separately by  $V_v$  and exported-mesh truth.
- SU2 is a coarse 2-D Euler calibration and must report convergence separately from successful execution.
- TACS and OAS use different wingbox idealizations; their mass/stress ratio is calibration evidence only.

Full interactive evidence, exact stage values, and configuration: <report.html>